

SOFTWARE ENGINEERING LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #44 – Developer Tools & IT Operations

System: Database Query Performance Profiler

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Section	5A
Course	Software Engineering Lab
Jira Project	My Scrum Space

Section 1: Agile Backlog Architecture (Epics & User Stories)

The functional requirements from Problem Statement #44 are converted into Agile Epics and User Stories. Each story is prioritized and estimated using Fibonacci-based story points.

Epic	Story ID	Actor	User Story	Priority	Story Points
Epic 1: Query Performance Analysis	FR001	Database Administrator	As a Database Administrator, I want to analyze PostgreSQL/MySQL EXPLAIN plans, identify sequential table scans on large tables, and receive index recommendations, so that I can identify inefficient query execution and improve database performance.	High	5
Epic 1: Query Performance Analysis	FR002	Database Administrator	As a Database Administrator, I want to ingest slow query logs and extract query text, execution time, frequency, and database source, so that query performance can be analyzed accurately.	High	3
Epic 1: Query Performance Analysis	FR003	Database Administrator	As a Database Administrator, I want the system to identify queries with high execution time or frequent occurrence, so that I can focus on queries that have the greatest performance impact.	High	3
Epic 2: Performance Reporting & Review	FR004	Backend Lead	As a Backend Lead, I want to receive a weekly performance optimization digest containing slow queries, detected performance issues, and recommended optimizations, so that I can review database performance trends and optimization opportunities.	Medium	3
Epic 2: Performance Reporting & Review	FR005	Database Administrator / Backend Lead	As a Database Administrator or Backend Lead, I want to view identified query performance issues and recommended optimization actions, so that I can make informed database optimization decisions.	Medium	2

Total Backlog: 16 Story Points across 2 Epics and 5 User Stories.

Jira Backlog Evidence

Spaces

My Scrum Space ...

Summary List Board **Backlog** Development Forms Timeline Docs Reports +

Search backlog AP Filter

Backlog (5 work items) 16 0 0 Create sprint

Create a whiteboard to plan your work Try

SCRUM-5 Execution Plan Analysis & Index Recommendation	Query Performanc...	To Do	5	^	
SCRUM-6 Slow Query Log Ingestion	Query Performanc...	To Do	3	^	
SCRUM-7 Performance Optimization Candidate Identification	Query Performanc...	To Do	3	^	
SCRUM-8 Weekly Performance Optimization Digest	Performance Repo...	To Do	3	=	
SCRUM-9 Performance Issues & Recommendations Review	Performance Repo...	To Do	2	=	

+ Create

recommendations.

Sprint 1 – Planning evidence

Spaces

My Scrum Space ...

Summary List Board **Backlog** Development Forms Timeline Docs Reports +

Search backlog AP Filter

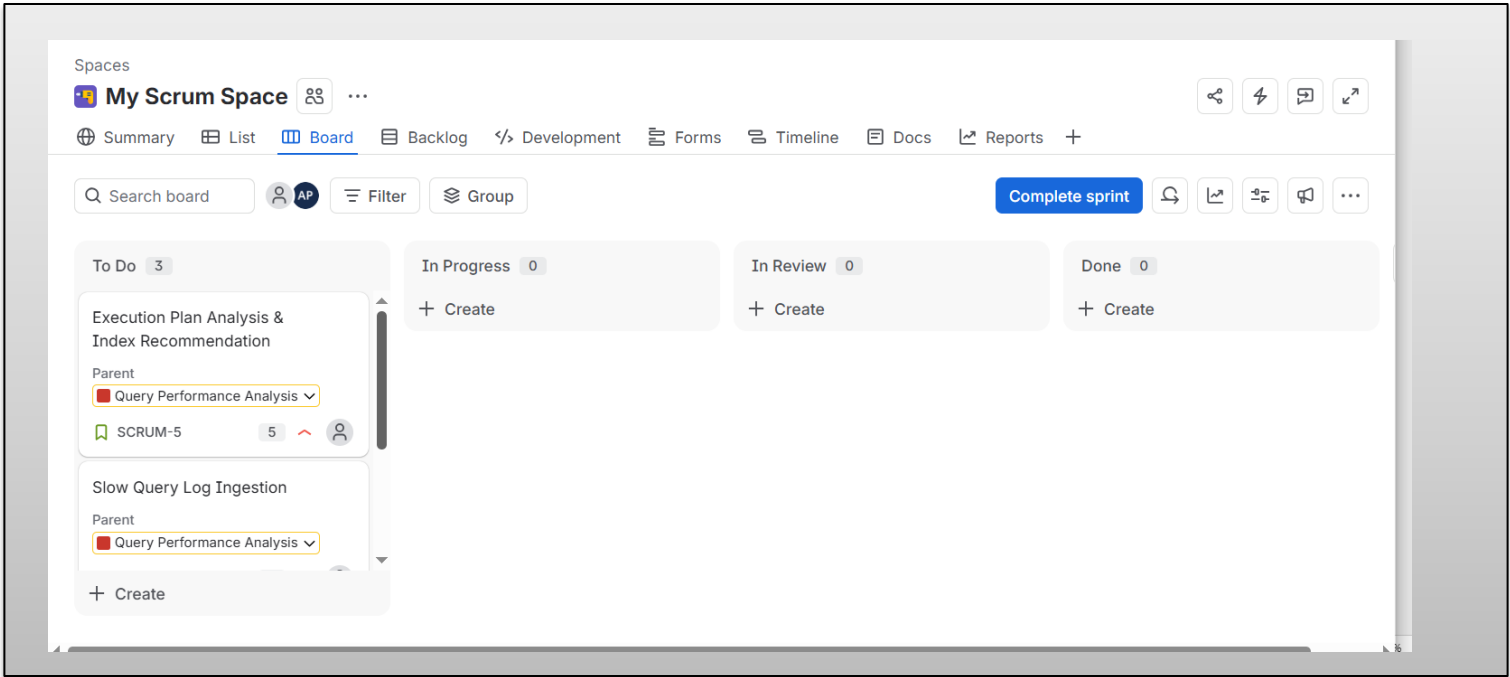
SCRUM Sprint 1 Add dates (3 work items) 11 0 0 Start sprint ...

SCRUM-5 Execution Plan Analysis & Index Recommendation	Query Performanc...	To Do	5	^	
SCRUM-6 Slow Query Log Ingestion	Query Performanc...	To Do	3	^	
SCRUM-7 Performance Optimization Candidate Identification	Query Performanc...	To Do	3	^	

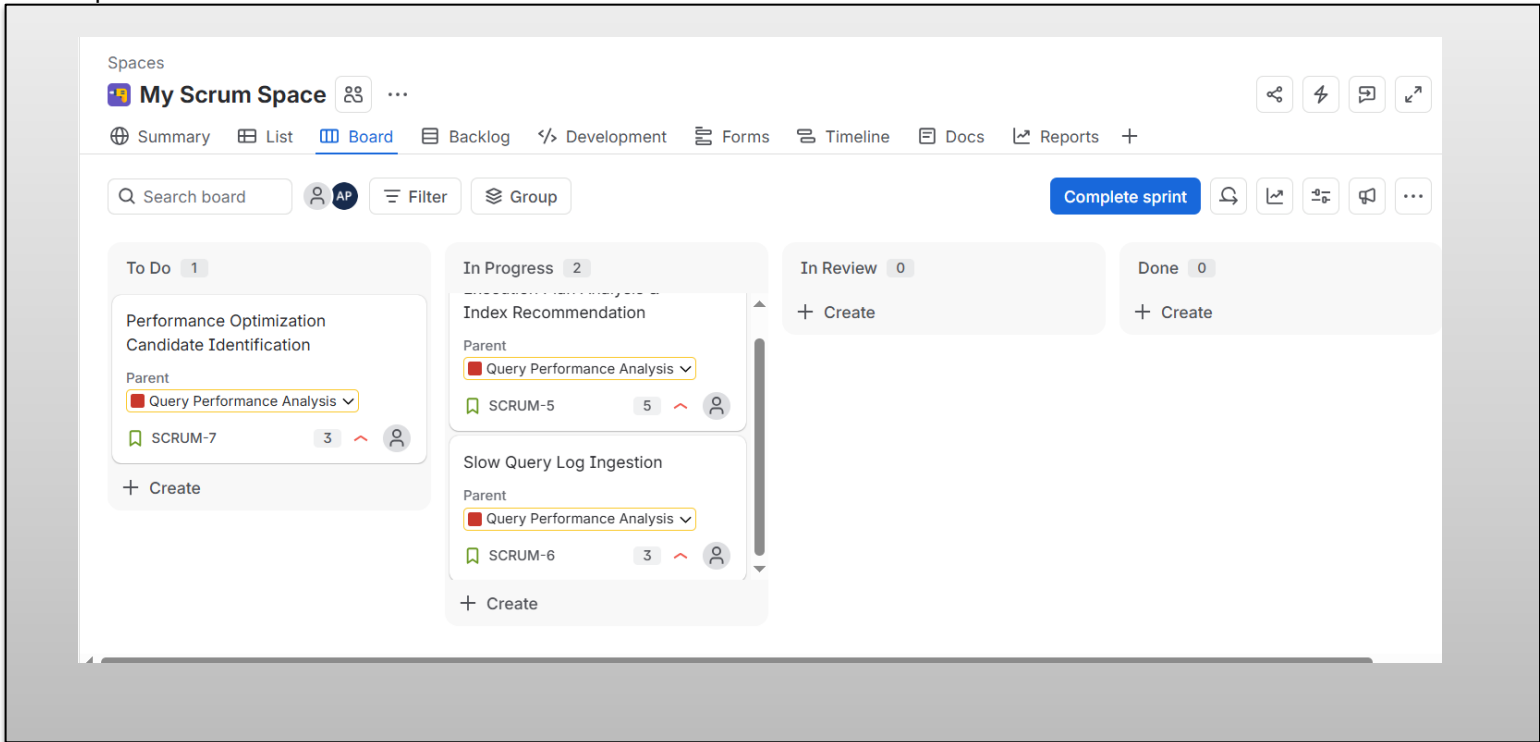
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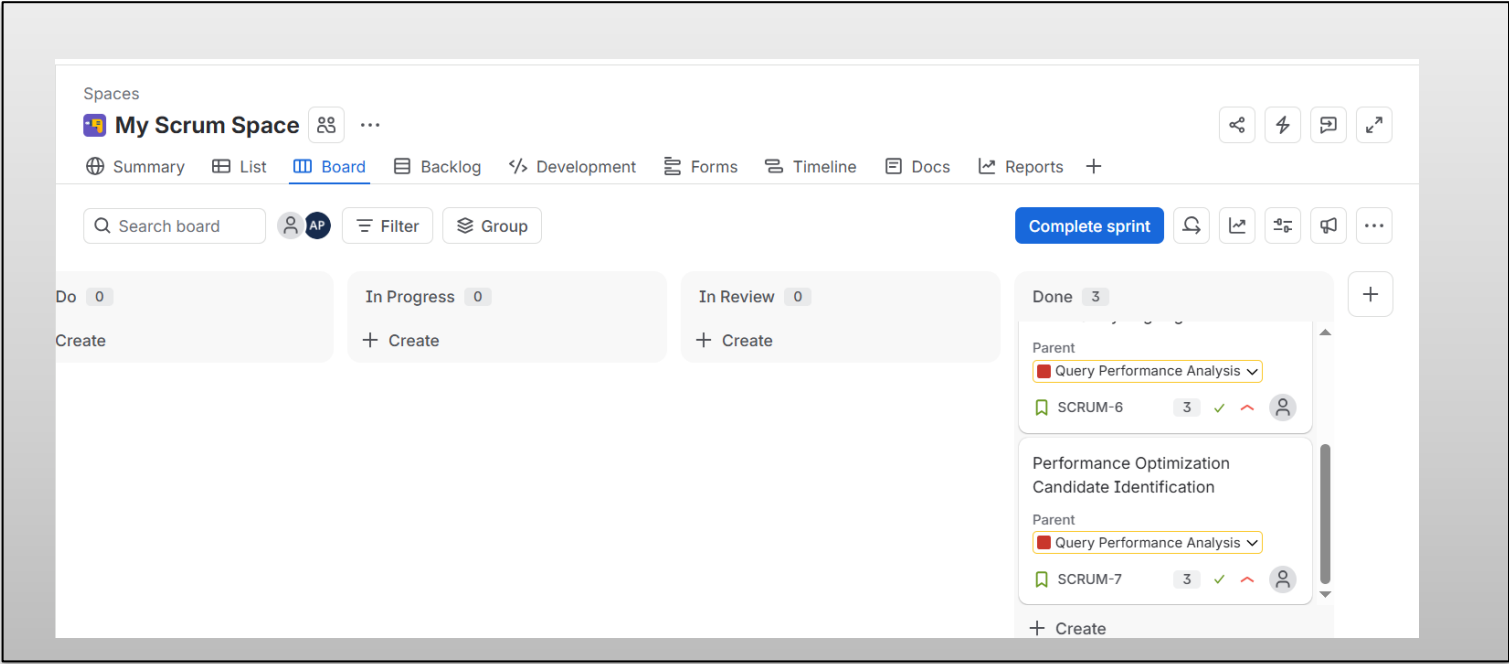
3 of 3 work items visible | Estimate: 11 of 11

Active Sprint1 initial Evidence

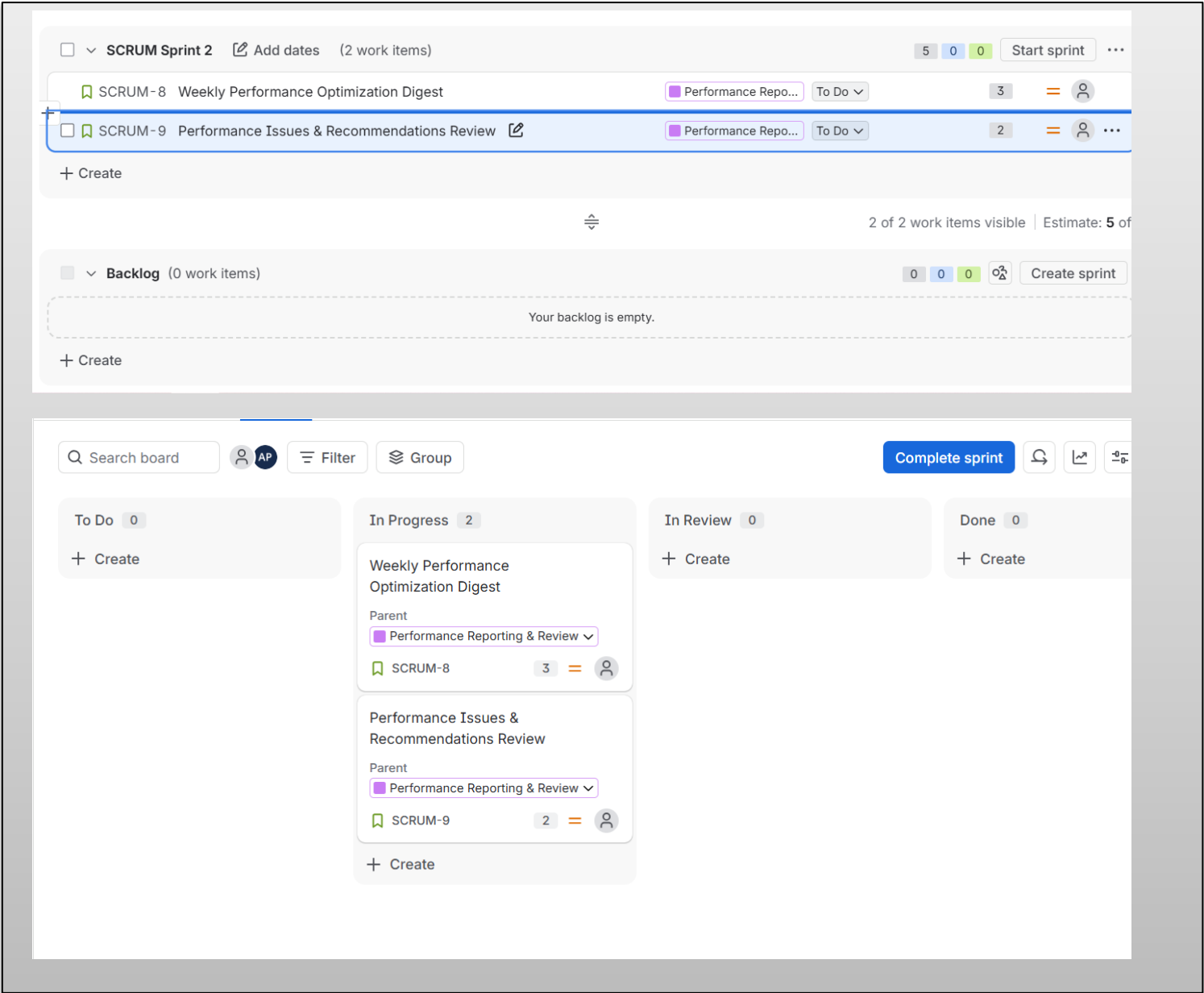


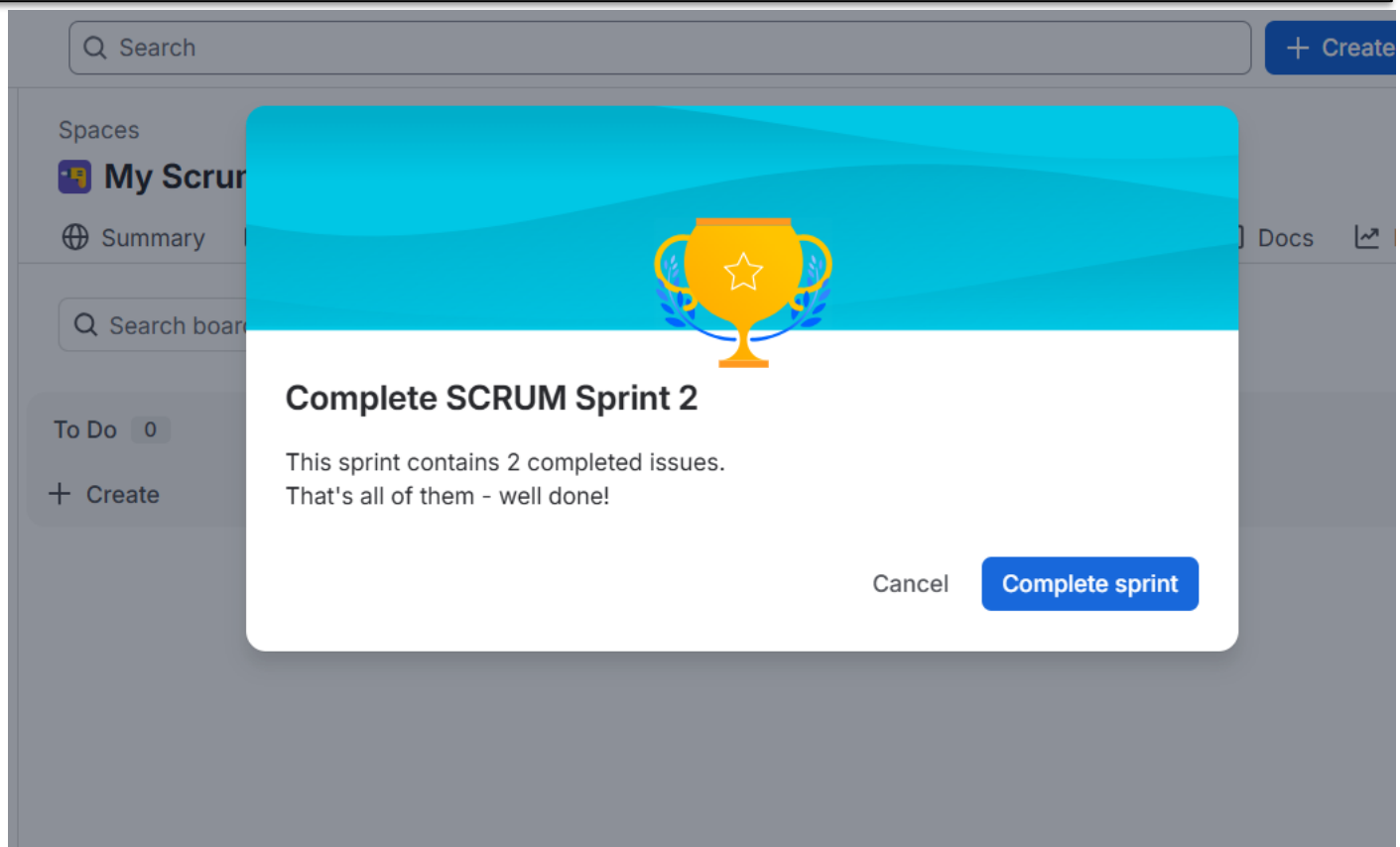
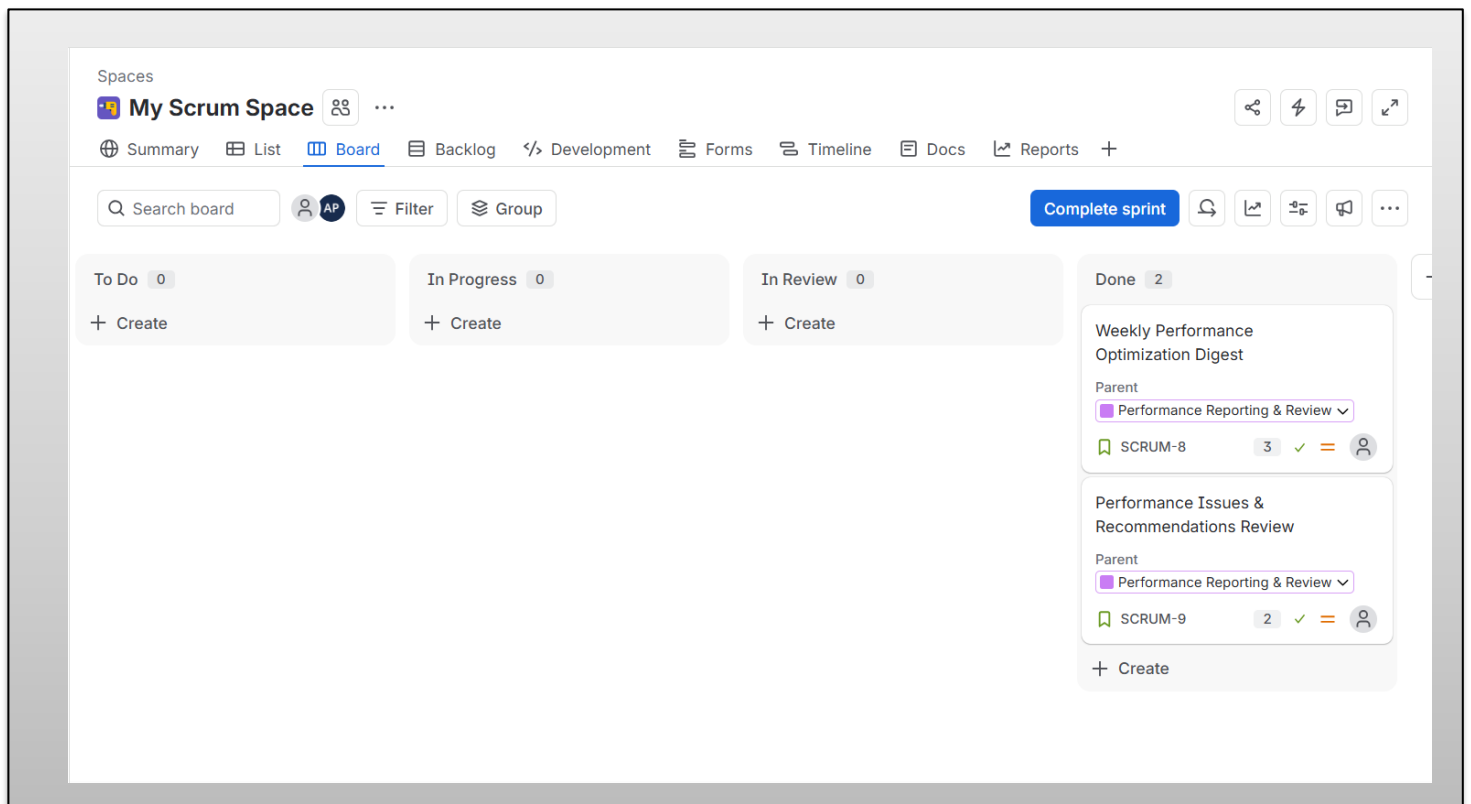
Sprint1 Execution evidence





Sprint 2 – Completed Sprint / Board Evidence

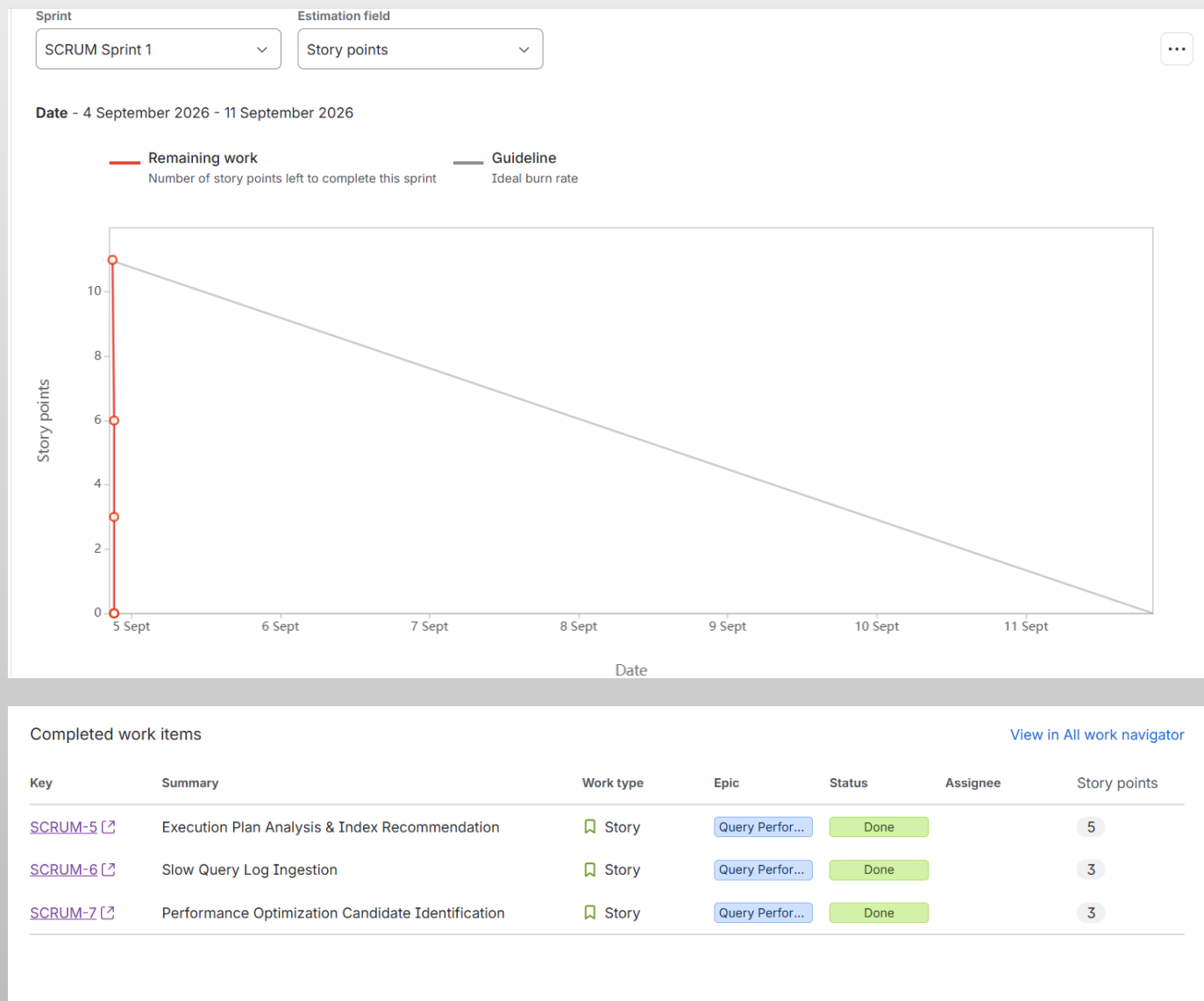




Section 3: Sprint Burndown & Metrics Analysis

Use the Jira Reports → Burndown Chart view after completing the sprint simulation. The analysis should describe the ideal guideline, remaining story points, completion pattern, and any deviations observed.

Sprint 1 – burndown chart

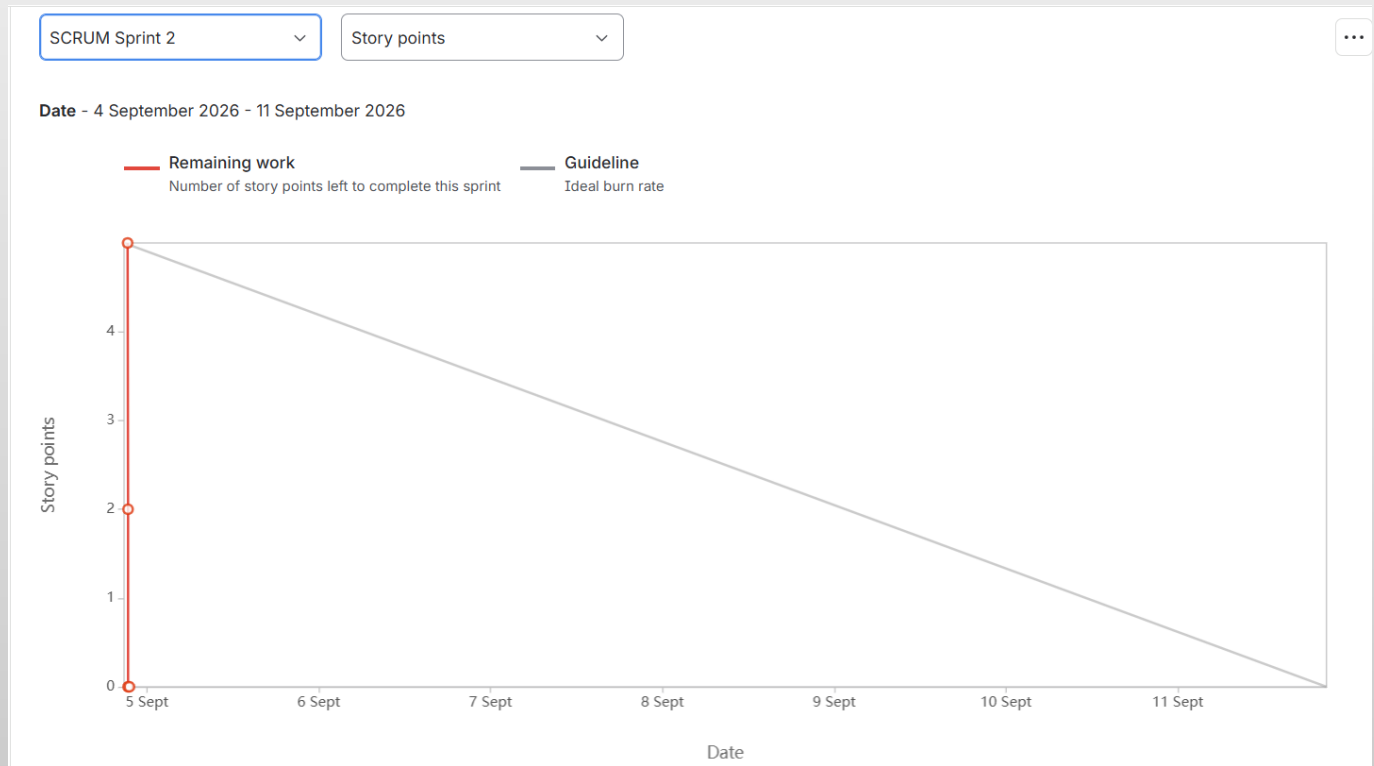


Burndown Analysis:

Sprint 1 started with a total of **11 committed story points** across three work items. The sprint included SCRUM-5 (5 points), SCRUM-6 (3 points), and SCRUM-7 (3 points). All three work items were completed, resulting in **0 remaining story points**. The actual remaining-work line shows a sharp drop from 11 to 0 at the beginning of the sprint.

The actual burndown differs significantly from the ideal guideline because the committed work was completed much earlier than the evenly distributed ideal burn rate. There were no remaining story points or visible delays by the end of the sprint. This indicates that the Sprint 1 work items were completed ahead of the ideal schedule.

Sprint 2- burndown Chart



Completed work items

[View in All work navigator](#)

Key	Summary	Work type	Epic	Status	Assignee	Story points
SCRUM-8	Weekly Performance Optimization Digest	Story	Performance ...	Done		3
SCRUM-9	Performance Issues & Recommendations Review	Story	Performance ...	Done		2

Burndown Analysis:

Sprint 2 started with **5 committed story points**. The burndown chart shows the remaining work dropping from **5 story points to 0** at the beginning of the sprint. The ideal guideline decreases gradually from 5 to 0 across the sprint duration.

The actual burndown therefore shows a **sharp early completion** compared with the ideal guideline. All committed work appears to have been completed much earlier than the planned burn rate, leaving **0 remaining story points** for the rest of the sprint. No visible delays or gradual completion pattern are observed in the chart.

Section 4: Reflection & Analysis

Question 1: Did your estimations reflect the actual effort?

The estimations generally reflected the actual effort observed during the simulation. The stories were estimated as **5, 3, 3, 3, and 2 story points**, with higher-point stories representing relatively larger or more complex tasks. All committed stories in the completed sprints were finished successfully, suggesting that the estimates were reasonably realistic for the simulated workload. The burndown charts also showed that the planned work could be completed within the sprint duration.

Question 2: Was your backlog well-prioritized?

Yes, the backlog was well-prioritized. The **High-priority query-analysis and performance-related stories** were scheduled before the **Medium-priority reporting and review stories**. This ordering supported the project workflow because the core analysis and optimization activities were addressed first, providing the necessary foundation for the later reporting and review activities. The prioritization also helped ensure that the most important performance-related work was completed early.

Question 3: How did your simulated sprint align with your plan?

The simulated sprints aligned well with the original plan. In **Sprint 1**, all three committed stories totaling **11 story points** were completed. In **Sprint 2**, the two committed stories totaling **5 story points** were also completed. There were no visible scope changes, and the committed work reached zero remaining story points in both sprints. Therefore, the sprint execution successfully met the planned sprint goals.

Question 4: What insights did the burndown chart give about your team's capacity?

The burndown charts showed that the team was able to complete its committed workload successfully. Sprint 1 had a capacity of 11 story points, while Sprint 2 had 5 story points, and both reached zero remaining work. The actual completion pattern showed a sharp reduction in remaining work compared with the ideal guideline, indicating that the simulated team completed the assigned work earlier than the ideal burn rate. These results suggest that future sprint planning should consider the team's demonstrated ability to complete the planned workload while avoiding unnecessary overcommitment.